Data Solution Design - DDE Aurora

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Document Details

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Introduction

This document describes the data design aspect of DDE Aurora. The project Aurora is meant to enhance the current functionality of the application with the addition of new components that will result in the following benefits for CBU and VB:

- Automation of the remaining deal types for CBU and VB, thereby eliminating all manual deal creation *(excluding Prepaid deals)*
  - Non-VSP deals
  - Trade-in deals (applicable to VB bulk deals as well)
  - On-biller deals
  - Customer specific deals (including deal match deals)
- Additional flexibility to configure device specific deals by enhancing the current business logic for headline, channel and campaign deals.
- Improve channel management capabilities by building a sub-channel deal creation layer which enables the Trade Marketing team to amend deals and add channel support as negotiated with specific channels at a granular level within the channel hierarchy.
- Initiate the consolidation of dealer qualifications management and maintenance within DDE and dealer channel hierarchy, and to optimize and improve the dealer_id management process.
- Develop functionality that enables the Trade marketing team to manage, update and reload deals post creation (e.g., deal extension and expiry)

These enhanced capabilities will bring further Dopa efficiencies but also create the foundation for the move to personalised deals.

> ⚠ **Disclaimer**: all details below in this document is subject to change as solution requirements evolve in the current Aurora Scoping sessions.
>
> The data solution design is dependent on how data is used within the application, so as user journeys and application queries mature, so too will the data design.

Scope of Requirements

The **Scope of Requirements** defines key business requirements that will shape the data design for DDE. It informs, the functional capabilities DDE must deliver, and the technical limitations that must be considered. These requirements are essential to ensuring that the design supports both current and future needs, maintains high performance under operational conditions, and aligns with business needs. Each requirement is meant to address the current pain points with the application.

Business Requirements

The following business requirements have been identified for DDE in the aurora scoping sessions and will be used inform the data design solution:

BR1 - Amendments (VAS, Differential and Discount amendments)

The current DDE state only allows for configurations on VAS's, Differentials and Discounts to be made prior to the headline deal creation on tariff level. The new requirement will include functionality that allows VAS, Differentials and Discounts to be updated per device and tariff combination post headline deal creation, the configuration includes

- Amend of VAS's (add, update & remove)
- Configuring of Differentials (updating a new differential amount)
- Update of Discounts (updating a new discount amount)
- Standard VAS's to be added to Data Tariffs with the ability to Amend the VAS's
- Adding Embedded VAS's on a deal

BR2 - Mid Cycle Deals (Custom Configurations for Device Launches)

DDE in its current state provides for one Main Cycle Deals to be generated per month with set validity dates. Flexibility is required to conduct multiple mini cycles on an ad hoc basis to accommodate launches of new devices that does not fall within the timeframe of standard deal cycles.

BR3 - Deal Term Configuration & Delete Enhancements

DDE in its current state allows for

1. Deal terms for 24 and 36 months, additional deal terms are required (12- and 48-month deal terms)
2. Deletion of one deal card at a time, enhanced deal deletion functionality is required with the ability to restore a deal card.

Requirement:

1. Include additional deal terms for 12 and 48 months.
2. Deletion and removal of multiple deal cards at a time
3. Removing deal cards from a specific cycle 4. Restoring of deal cards

Functional Requirements

The functional requirements outline the specific capabilities DDE must support:

FR1 - Amendments (VAS, Differential and Discount amendments)

The DDE application must enable amendments of VASes at headline.

FR2 - Mid Cycle Deals (Custom Configurations for Device Launches)

The solution should provide the user with the ability to create an additional deal cycle that is linked to a selected price list. Functionality to do custom configurations before a deal cycle starts for specific devices by selecting devices, determine starting tariffs, adding custom differentials, determine applicable VAS. Ability to configure for 1 device or for multiple devices at once. Deal configurations section should be split into standard configurations and custom configurations.

FR3 - Deal Term Configuration & Delete Enhancements

The user needs to have the option of adding new deal terms.

- **FR4**: Automation of No-VSP deals to eliminate all manual deal creation of this deal type.

Data Solution Design

The Data Solution section of this document outlines the strategic approach to managing, storing, and structuring data for DDE. With MySQL serving as the primary backend database, the focus is on delivering a scalable, efficient, and secure data environment. This section provides detailed insights into the design principles, the data models that structure the application's data, and the storage solutions implemented to ensure optimal performance and reliability.

The goal of the Data Solution Design is to ensure that the application’s data infrastructure aligns with business needs, supports high availability, and meets future scalability requirements. The subsections that follow will delve into the specific technical choices made for data modeling and storage, detailing how these solutions contribute to the overall efficiency and functionality of the system.

Subsequent sections will elaborate on:

- **4.1.1 Data Models**: The logical representation of the database structure, defining how data is organized, related, and maintained.

- **4.1.2 Data Storage Solutions**: The physical storage strategies, including considerations for performance, security, and capacity management.

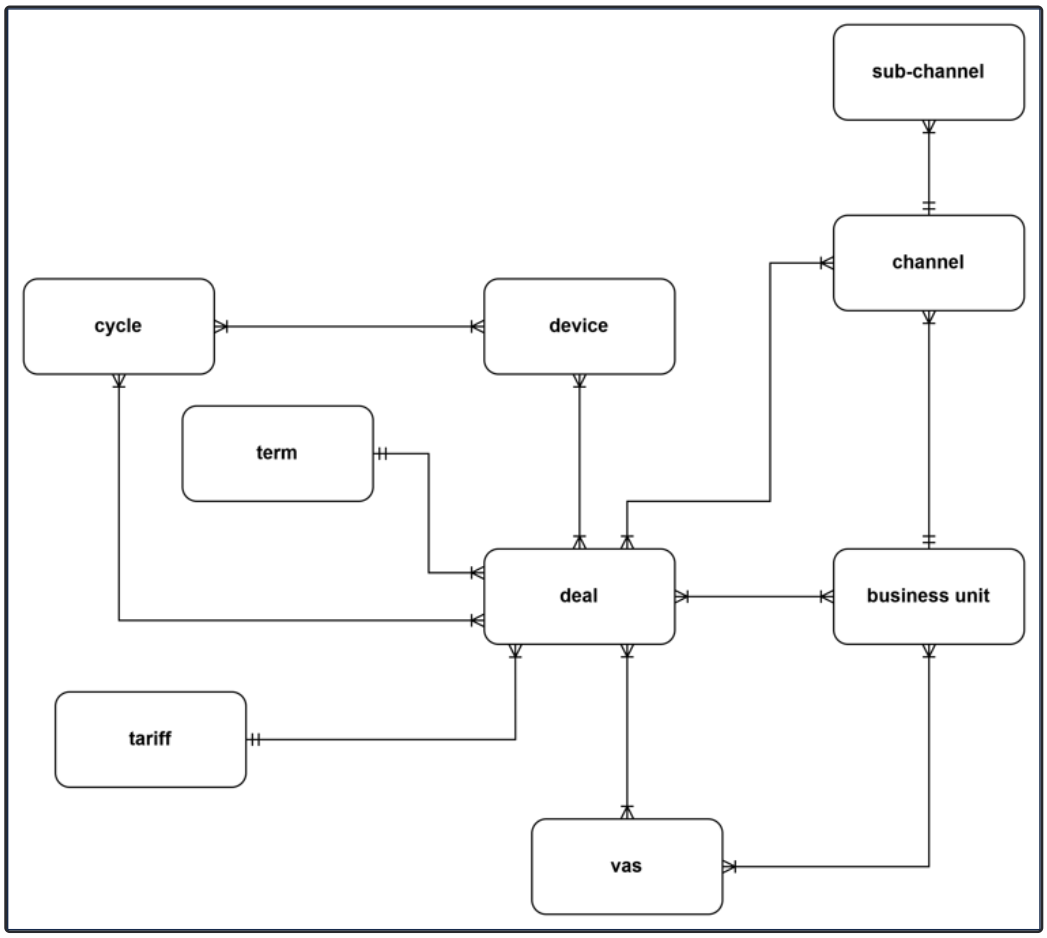
Data Architecture

Data Models

The data models presented in this section provide a comprehensive representation of the DDE's future data structure, designed to meet the business and functional requirements defined earlier. This section includes the following subsections:

Conceptual Data Model

The conceptual data model/data taxonomy presents a high-level view of the core entities involved in DDE and their relationships. This model abstracts the detailed specifics and focuses on the fundamental structure that supports the key business processes. The entities here align directly with the business requirements, ensuring that each core element is mapped to a major business need. Relationships between these entities ensure data cohesion and an accurate reflection of real-world interactions.



DDE’s Conception Data Model

The main entities include:

- **device**: Represents a device(s) which is part of the deal.
- **tariff**: Represents various services offered by Vodacom to customers on their network.
- **deal**: Represents the contractual agreements between customers and Vodacom.
- **VAS**: Value added services as part of the deal, potentially to increase uptake of Vodacom contracts.
- **term**: Contractual term of the deal.
- **cycle**: Represents the validity period of a deal.
- **channel**: Represents channels where deals are sold.
- **business unit**: Represents the business unit where deals are sold.

The Data Taxonomy

The data taxonomy presents a high-level view of the core entities involved in DDE and their relationships. This model was developed at Condorgreen and helps abstracts the detailed specifics of related entities and focuses on the fundamental structure that supports the key business processes. It also adds a lot of information from a typical conceptual model such as separating and inferring relationships between the entities.

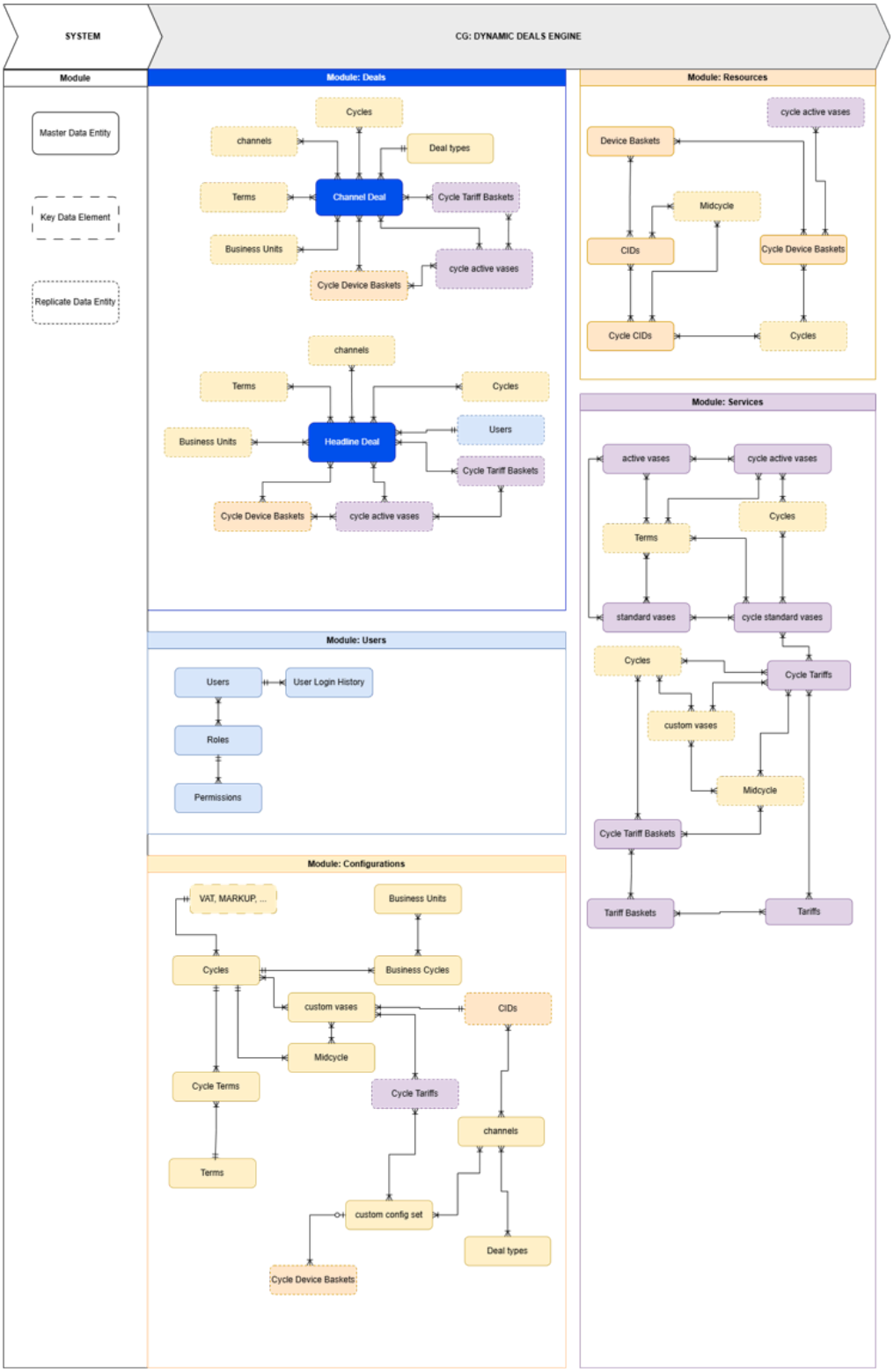


Figure 2: Data Taxonomy

Logical & Physical Data Model

Building on the conceptual data model, the logical data model refines the representation to include more details about attributes, relationships, and data types without considering

implementation specifics. It breaks down each entity into its key fields, describing the properties and the relationships between entities in a more structured way. The logical data model bridges the gap between business requirements and physical implementation. This detailed representation ensures that DDE is capable of fulfilling all functional and business requirements.

FR1 - Amendments (VAS, Differential and Discount amendments)

In order to cater for this specification, the `enr_cycle_standard_vases` table needed to be modified so that we could capture device-tariff and VAS mapping by adding a `ccs_id` column. This will allow us to modify the vases per device-tariff mapping.

In order to cater for the specification for custom configurations, the `enr_cycle_device_baskets` and `enr_cycle_tariffs` needed to be modified to include the `ccs_id` columns. An additional column was added in the `enr_cycle_device_baskets` table to capture the channel component as well as certain custom configs could apply to certain channels and not others. An additional table is needed to be created to keep track of the custom configs. The name of the table is `cfg_custom_config_set`.

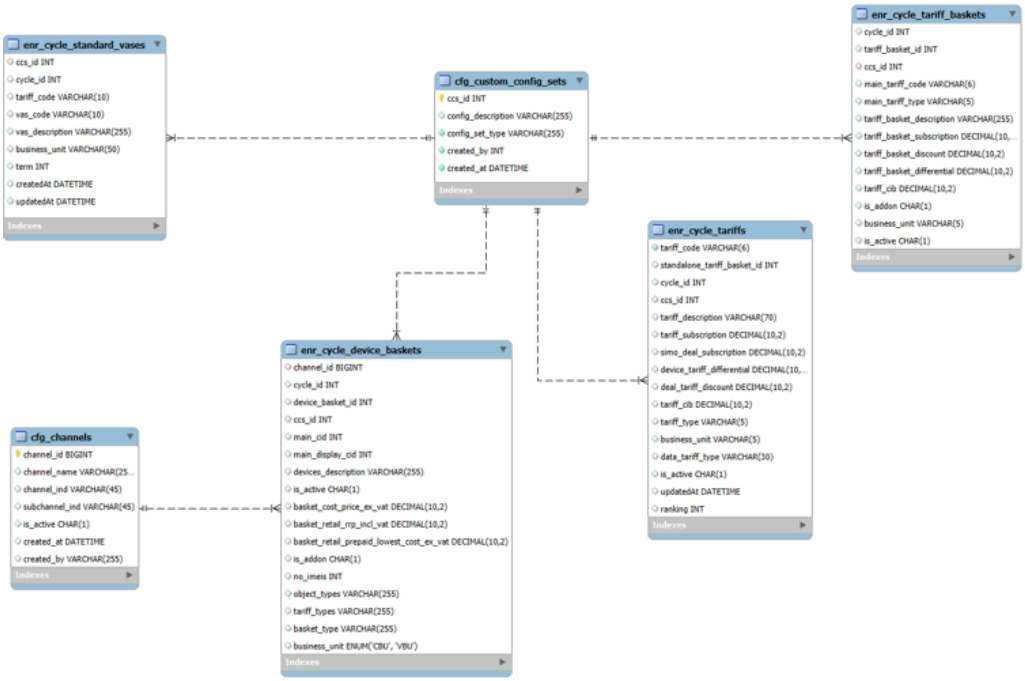


Figure 3: Custom Configs Schema Design

FR2 - Mid Cycle Deals (Custom Configurations for Device Launches)

In order to cater for this specification, the `enr_campaigns` table is repurposed so that it can keep track of `midcycle_name` / `campaign_name` along with the begin and end date of the new product launch cycle's deals. It has been altered by adding extra columns to keep an audit trail of who actually created/updated a midcycle. All cycles will be considered a midcycle be it Generic or device launch. This design makes makes it easier to not only keep track of midcycles but also maintains the already existing structures built into the system. The proposed schema design for this is show in the figure below. The solution should provide the user with the ability to create a midcycle that is linked to a selected price list.

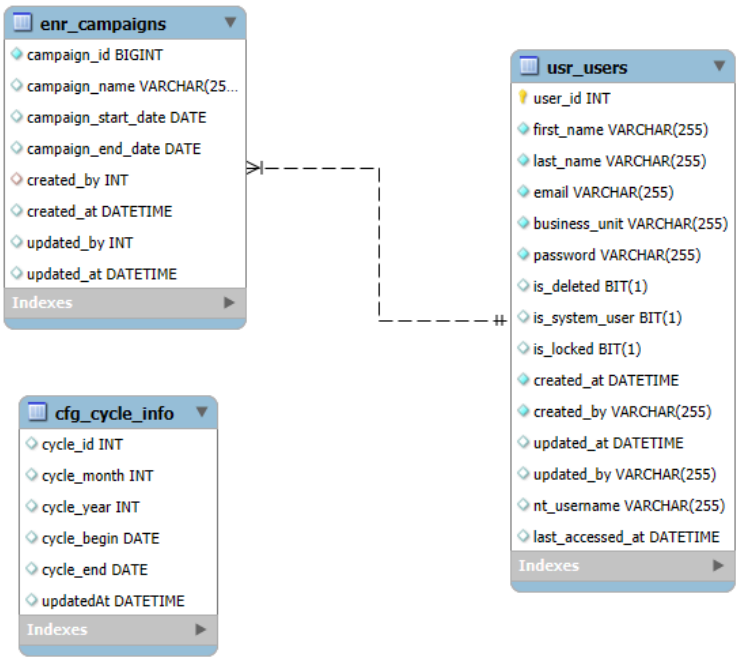


Figure 4: Proposed Schema for Midcycles

FR3 - Deal Term Configuration & Delete Enhancements

No change was made to the current schema as it can capture the new requirements for additional deal terms. The term specific parameters are stored in the `cfg_deal_config_hist` table which is cyclic. It allows users to configure term specific parameters per cycle.

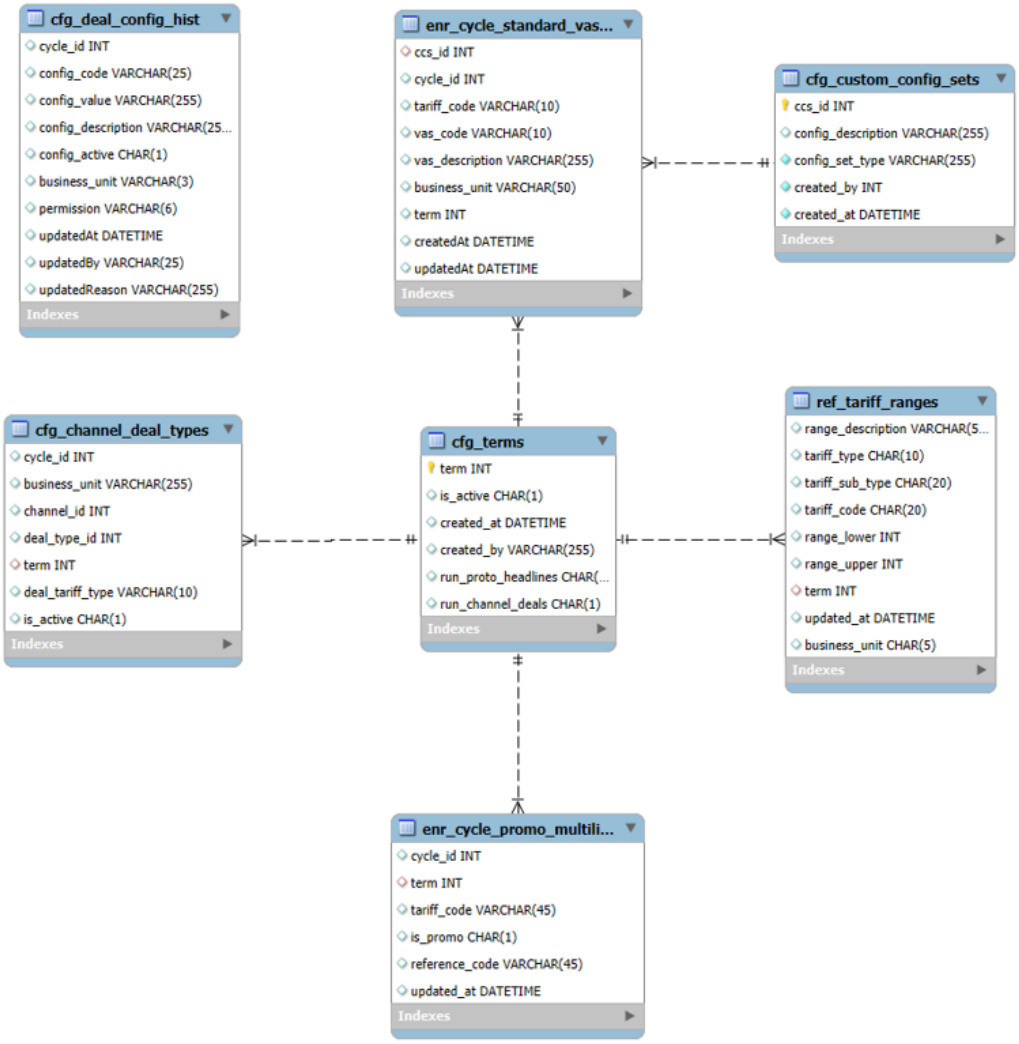


Figure 5: Deal Term Config Proposed Schema

Relationships

Data Taxonomy & Relative Cardinalities

Data Storage Solutions

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